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KAPNET CORPUS SYNTHESIS: A STANDALONE LITERATURE REVIEW AND INTERNAL DISSERTATION ON THE DEVELOPMENT OF A BITCOIN-NATIVE PRE-SETTLEMENT COORDINATION SUBSTRATE

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Abstract. This dissertation synthesizes the current Kapnet corpus as an internal literature review. Its purpose is neither merely archival nor promotional. It aims to identify what has already been established, what conceptual strata have stabilized, what tensions remain unresolved, and how the corpus should now be condensed into a coherent canon. The Kapnet literature has evolved from a broad systems-theoretic reinterpretation of Bitcoin toward a progressively narrower protocol program centered on weakwork, braids, knots, bounded deterministic replay, current-Bitcoin-compatible anchoring, treasury accounting, and institutional state.

The central thesis of the corpus is stable: Bitcoin remains the final proof-of-work settlement layer, while Kapnet prices, structures, replays, and selectively anchors coordination that occurs before settlement. The dissertation argues that the corpus has already succeeded in defining a coherent and distinctive design space: a Bitcoin-native coordination substrate whose canonical objects are submissions, braids, knots, and anchors, and whose primary novelty is the pricing and replay of institutional coordination before final settlement. At the same time, the review shows that the corpus still lacks a frozen reference specification, a working implementation, a complete adversarial model, and an empirical benchmark layer. The result is a comprehensive internal map of the corpus as it stands, together with a canonical ordering for future refinement.

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1. Introduction

The Kapnet corpus is now large enough that it requires its own literature review. This is a sign of progress, but also of danger. Progress, because a set of ideas has persisted long enough to generate recurring objects, recurring distinctions, recurring problems, and recurring technical language. Danger, because without synthesis the corpus risks collapsing into self-overlap, duplication, and conceptual drift.

The present dissertation is therefore written as an internal review of the corpus itself. It does not ask whether Kapnet has already been accepted by outside institutions, implemented in production, or standardized beyond the drafting stage. It asks a prior question: what has the corpus, taken on its own terms, actually achieved? Which concepts have become stable? Which layers remain speculative? Which papers are foundational, which are transitional, and which are now candidates for canonical status?

The answer may be stated in compact form. The corpus has progressed through four phases:

1. Bitcoin as extensible consensus state;
2. Kapnet as pre-settlement coordination market;
3. Kapnet as a multi-agent political economy;
4. Kapnet as a candidate machine.

That progression is real. It does not yet amount to a finished protocol, but it does amount to a coherent research program with a distinctive object.

2. Method and Scope

Definition 2.1 (Corpus)**.** The *Kapnet corpus* is the set of draft papers, annexes, kernel specifications, internal summaries, and formalizations produced around the following recurring themes:

1. Bitcoin as an extensible proof-of-work consensus state;
2. weakwork, braids, replayability, knots, and anchors;

3. guilds, rooms, submissions, TXXM, and ASC;
4. treasury accounting, unit economy, and agent order;
5. KapVM-Core, byte schemas, opcodes, witnesses, and test vectors;
6. recent annexes on plugin ecology, agent swarms, daemon archetypes, and institutional history.

Remark 2.2. This dissertation is an internal literature review rather than a neutral history written from outside the project. Its standard of judgment is therefore primarily one of conceptual coherence, structural maturity, and protocol usefulness rather than publication prestige.

Definition 2.3 (Canonicality criterion)**.** A concept, paper, or formal object in the corpus is called *canonical* if it satisfies all of the following:

1. it recurs across multiple documents;
2. later drafts sharpen rather than replace it;
3. it supports or constrains other concepts in the corpus;
4. its removal would leave a structural hole.

3. General Trajectory of the Corpus

3.1. **From broad philosophical frame to protocol kernel.** The earliest layer of the corpus is philosophical and systems-theoretic. Bitcoin is reinterpreted not only as money or payment network, but as an extensible proof-of-work consensus state with scarce blockspace, deterministic settlement semantics, and path-dependent evolution. This stage establishes the base substrate.

The second stage introduces overlay-native objects. Weakwork becomes a priced admission resource. Braids become weighted partially ordered candidate histories. Replayability becomes a condition of canonicality. Knots become checkpoint objects. Anchors become compressed settlement interfaces into Bitcoin.

The third stage enlarges the frame into a political economy. Hashproducers, weakwork brokers, patient capital providers, DAO operators, utility providers, and settlement agents are differentiated. Hedlbits, satdepth, adjusted satdepth, and hodlyield appear as a layered unit ontology. Treasury continuity becomes as important as replay correctness.

The fourth stage begins to close the machine gap. KapVM-Core becomes a bounded deterministic replay model. Typed fragments are separated into commutative, ordered, and commitment classes. Byte schemas, opcode families, error codes, witnesses, archive receipts, and replay test vectors appear. At this point the corpus begins to transition from theory into specification.

3.2. **A compressed developmental table.**

Phase	Main question	Principal objects
Foundational	What is Bitcoin as a system?	UTXO state, blockspace, extensible node utility
Market-theoretic	What is Kapnet as an overlay?	weakwork, braids, replayability, knots, anchors
Political-economic	Who produces and consumes coordination?	agent hierarchy, treasury continuity, multi-unit economy
Protocol-kernel	What exact machine should run?	KapVM-Core, byte schemas, opcodes, witnesses, test vectors

4. Foundational Layer: Bitcoin as Extensible Consensus State

4.1. **Main contribution.** The foundational text of the corpus is the paper tentatively titled *Bitcoin: A Proof-of-Work Extensible Consensus State*. Its main achievement is to formalize Bitcoin as:

1. a distributed timestamp server,
2. a deterministic UTXO transition system,
3. a scarce blockspace market,
4. a path-dependent consensus substrate,
5. a system whose node utility may be extended without immediately mutating base validity.

4.2. What this layer gets right.

Proposition 4.1 (Foundational strength)**.** *The foundational paper correctly separates consensus-validity, local policy, and higher-order interpretation as distinct layers.*

Proof. The paper consistently distinguishes between base consensus rules, relay and implementation choices, and meta-interpretive or overlay uses of valid data. This separation later becomes essential when introducing Kapnet without claiming it alters Bitcoin consensus. □

Remark 4.2. The phrase *extensible node utility* is one of the strongest contributions of this layer. It prepares the way for Kapnet by showing that nodes can do more than validate blocks without ceasing to be Bitcoin-valid nodes.

4.3. Limitations of the foundational layer.

Remark 4.3. At this stage Kapnet appears mainly as implication rather than machine. The foundational paper opens the design space but does not yet occupy it with a fully specified overlay. That is appropriate for its role, but it also marks its limit.

5. Market-Theoretic Layer: Weakwork, Braids, Knots, and Anchors

5.1. **Main contribution.** The second core document, *Bitcoin-Adjacent Coordination as a Weighted Pre-Settlement Market: Weakwork, Braids, and Mining Implications*, is the first paper in the corpus where Kapnet becomes a genuinely autonomous protocol object. Its major contributions are:

1. weakwork as a locally verifiable priced admission resource;
2. admissible message graphs;
3. braids as weighted candidate histories;
4. replayability as a precondition for canonicality;
5. knots as checkpoint objects;
6. anchor commitments as compressed settlement interfaces;
7. two-stage market decomposition between relay-side and settlement-side pricing.

5.2. **Distinctive conceptual achievement.**

Definition 5.1 (Pre-settlement coordination market)**.** A *pre-settlement coordination market* is a protocol layer in which scarce rights to admit, order, replay, and preserve coordination events are priced prior to final settlement.

Remark 5.2. This concept is arguably the central intellectual novelty of the entire corpus. Most Bitcoin-adjacent systems focus on settlement, execution, privacy, or throughput. Kapnet instead foregrounds the scarcity and pricing of coordination before settlement.

5.3. **The braid concept.**

Definition 5.3 (Braid)**.** A *braid* is a weighted partially ordered candidate history rather than a mere unordered pool of visible messages.

Proposition 5.4 (Braid distinctiveness)**.** *The braid is more conceptually powerful than a mempool analogy because it moves the unit of competition from isolated objects to candidate histories.*

Proof. An isolated-message model can price visibility but cannot natively price structured candidate order. A braid includes dependency structure, replayability conditions, weight terms, and future checkpoint significance. It therefore captures history competition rather than queue occupancy. □

5.4. **Key strength and key weakness.**

Remark 5.5. The key strength of this layer is that it provides the first serious formal language for the overlay. The key weakness is that some theorems remain dependent on axioms whose machine realization had not yet been specified at the time: bounded replay, path-independence on useful fragments, concrete anchor formats, and proof-bearing client semantics.

6. Political-Economic Layer: Agents, Treasuries, and Multi-Unit Scarcity

6.1. **Main contribution.** The next major layer of the corpus introduces a more complex economic ontology. Kapnet is no longer treated as an overlay with generic users, but as a hierarchy of productive and consuming roles. The principal agent classes are:

$$\begin{aligned}
\mathcal{A}_0 &:= \text{hashproducers},\\
\mathcal{A}_1 &:= \text{weakwork allocators and service-credit markets},\\
\mathcal{A}_2 &:= \text{patient capital and hedlbit providers},\\
\mathcal{A}_3 &:= \text{Kapnetizens},\\
\mathcal{A}_4 &:= \text{DAO operators and treasury stewards},\\
\mathcal{A}_5 &:= \text{utility providers},\\
\mathcal{A}_6 &:= \text{settlement and anchor agents}.
\end{aligned}$$

6.2. **Kapnet as stacked economy of agency.**

Proposition 6.1 (Agent-order contribution)**.** *The corpus advances beyond flat network talk by presenting Kapnet as a stacked economy of agency in which different classes govern different scarcities.*

Proof. Hashproducers govern industrial scarcity; weakwork allocators govern admission-liquidity scarcity; patient capital providers govern time-backed reserve scarcity; DAO operators govern continuity; utility providers govern service delivery; settlement agents govern Bitcoin interface. These are structurally distinct functions rather than one generic user class. □

6.3. **The economy of units.**

Definition 6.2 (Principal unit family)**.** The recurring unit family of the corpus is

$$\mathfrak{U} = \{\mathrm{sat}, \mathrm{B}, \mathrm{WU}, \mathrm{H/s}, \mathrm{J}, \mathrm{W}, \mathrm{WW}, \mathrm{HB}, \mathrm{SD}, \mathrm{ASD}, \mathrm{HY}\}.$$

Remark 6.3. This unit system is one of the most ambitious and unusual parts of the corpus. It tries to model not only money and message size, but industrial production, patience, reserve depth, and reward entitlement.

Proposition 6.4 (Multi-scarcity insight)**.** *The corpus correctly treats Kapnet as a multi-scarcity system rather than a fee-only system.*

Proof. Satoshis measure final monetary settlement. Hashrate and joules measure industrial production. Weakwork measures admission and ranking effort. Hedlbits measure patience-backed rights. Adjusted satdepth measures reserve quality through time. Hodlyield measures reward over temporal reserve states. Utility balances measure service continuity. These quantities arise from distinct production and depletion processes. □

6.4. **Limitation of this layer.**

Remark 6.5. The political-economic layer is conceptually rich but not yet fully algebraically closed. The units are named, interpreted, and related, but the earliest drafts do not yet embed them in a fully explicit state-update discipline with bounded issuance, sinks, and no-free-cycling constraints.

7. Meta-Layer: Open Problems, Roadmaps, and Research Discipline

7.1. **Main contribution.** A later group of documents turns the corpus back onto itself. These papers identify the central dependency order of unresolved work:

$$\text{execution} \to \text{anchors} \to \text{clients and proofs} \to \text{accounting and simulation.}$$

7.2. **The decisive reframing.**

Proposition 7.1 (Machine gap proposition)**.** *The roadmap layer correctly identifies that the corpus proves Kapnet is a coherent category of thing, but not yet a well-specified machine.*

Proof. Earlier papers define persistent objects and consistent distinctions, but do not yet freeze serialization, transition semantics, witness formats, client sync rules, or empirical benchmarks. Therefore the category is coherent while the machine remains in formation. □

Remark 7.2. This is one of the healthiest moments in the corpus. Rather than masking the gap, the project begins naming it explicitly and ordering the work required to close it.

8. Protocol-Kernel Layer: KapVM-Core and Byte-Level Specification

8.1. **Main contribution.** The strongest technical documents in the corpus are the KapVM-Core draft and its annexes. These begin to define:

1. a bounded deterministic replay machine;
2. typed activity classes;
3. commutative, ordered, and commitment fragments;
4. canonical serialization rules;
5. opcode families;
6. error-code tables;
7. knot witness structure;
8. archive receipts and fraud-proof bundles;
9. replay test vectors.

8.2. **Constitutional transition machine.**

Definition 8.1 (Constitutional transition machine)**.** A *constitutional transition machine* is a replay machine whose primary purpose is not arbitrary computation, but structured institutional state transition: room creation, guild amendment, membership changes, treasury operations, namespace control, dispute handling, and checkpointing.

Proposition 8.2 (Kernel orientation)**.** *One of the strongest advances of the corpus is the shift from imagining KapVM as a general VM to defining it as a constitutional transition machine.*

Proof. Narrowing the machine target reduces semantic ambiguity, simplifies boundedness proofs, makes governance and treasury state first-class, and aligns execution with the actual intended institutional objects of the protocol. □

8.3. **Commutative versus ordered fragments.**

Remark 8.3. Another decisive improvement is the split between commutative and ordered fragments. Rather than insisting that all activities be equally reorderable or equally sequenced, the corpus now recognizes that:

1. posts, votes, and validations often admit a commutative treatment;
2. treasury actions, namespace transfers, membership changes, and disputes do not.

This is a large gain in rigor.

8.4. **Kernel-layer weakness.**

Remark 8.4. Despite this progress, the kernel layer is still pre-implementation. It lacks frozen canonical fixtures, production byte dumps, a reference interpreter, and benchmark results. It is the strongest technical layer, but not yet finished.

9. Expansion Frontier: Recent Annexes and Conceptual Breakthroughs

9.1. **Submission centrality.**

Definition 9.1 (Submission)**.** A *submission* is the atomic coordination object of Kapnet: the canonical content-addressed unit from which discourse, validation, governance action, and executable state-facing interaction are built.

Proposition 9.2 (Submission centrality thesis)**.** *The corpus increasingly converges on the submission as the irreducible state-facing object of coordination.*

Proof. Later annexes repeatedly elevate the submission above generic message language. It becomes the center of hashing, dependency tracking, weighting, replay, archival storage, and checkpoint commitment. □

9.2. **ASC, rooms, and guilds.**

Definition 9.3 (Amethyst State Chain)**.** The *Amethyst State Chain* is the institutional state domain on which rooms, guilds, submissions, and related replay objects are organized.

Remark 9.4. A major refinement of the corpus is the move from treating rooms and guilds as merely sociological categories to treating them as replayable institutional state domains on ASC.

9.3. **TXXM and transport.**

Remark 9.5. The later corpus also sharpens the role of TXXM. The most coherent reading is:

1. submission = atomic state object;
2. TXXM = transport, routing, and bundling envelope for submissions and related coordination payloads.

This distinction should likely become canonical.

9.4. **Template economy and candidate institutional history.**

Definition 9.6 (Template economy)**.** A *template economy* is the market process by which candidate relay, replay, checkpoint, and anchor packages are assembled under multiple simultaneous constraints.

Definition 9.7 (Institutional braid)**.** An *institutional braid* is a candidate history whose replay affects ordered-fragment governance or treasury state rather than only discourse-level append-only state.

Remark 9.8. These later concepts are among the most fertile in the corpus. They extend the early braid idea from candidate visibility ordering to candidate institutional history.

9.5. **Kap-plets, plugins, agent swarms, and daemon archetypes.**

Remark 9.9. Recent annexes also move Kapnet toward an application and automation ecology:

1. kap-plets as bounded composable logic components;
2. plugin ecology rather than monolithic app logic;
3. agent-capable rooms and swarm coordination;
4. daemon archetypes as recurring automation patterns.

This material is conceptually powerful, though not all of it yet sits inside the frozen protocol kernel.

10. Core Concepts that Now Appear Canonical

This section identifies the concepts that now seem stable enough to count as canonical.

10.1. **Bitcoin as final settlement substrate.**

Summary 10.1. The corpus is unambiguous that Bitcoin remains the final proof-of-work settlement layer. Kapnet does not replace Bitcoin consensus.

10.2. **Weakwork as non-consensus priced admission.**

Summary 10.2. Weakwork is consistently treated as:

1. local rather than consensus-global,
2. verifiable,
3. costly,
4. upstream of settlement,
5. useful for admission and ranking.

10.3. **Braid as candidate history.**

Summary 10.3. The braid is now a canonical Kapnet object. Its importance lies in treating history competition, not merely queue competition, as the overlay's natural object.

10.4. **Knot as checkpoint object.**

Summary 10.4. The knot is canonically the selected checkpoint object: chosen braid, resulting state root, trace commitment, and optional anchor data.

10.5. **Replayability before canonicality.**

Summary 10.5. A candidate must be replayable before it can become canonical. This is one of the strongest and most stable rules in the corpus.

10.6. **KapVM as bounded deterministic machine.**

Summary 10.6. The corpus now strongly converges on a small typed replay machine rather than a broad or unrestricted smart-contract VM.

10.7. **Governance as ordered state.**

Summary 10.7. Rooms, guilds, treasury actions, membership, namespace rights, and disputes belong to the ordered state fragment.

10.8. **Kapnet as pre-settlement coordination market.**

Summary 10.8. This is the best one-line description of the full corpus: Kapnet is a Bitcoin-native pre-settlement coordination market.

11. Major Unresolved Tensions in the Corpus

A mature literature review must identify not only strengths but unresolved tensions.

11.1. **Broad vision versus narrow kernel.**

Remark 11.1. The corpus still oscillates between two registers:

1. a maximal civilizational and philosophical register;
2. a narrow protocol-specification register.

This is not a flaw in itself, but it requires stricter document layering. Visionary papers and kernel papers should no longer carry the same burden.

11.2. **Submission versus TXXM.**

Remark 11.2. There remains some ambiguity about whether the atomic object of Kapnet is the submission or the TXXM envelope. The strongest synthesis is that the submission should be the canonical state object, while TXXM should serve as transport and bundling envelope.

11.3. **Weakwork as scalar rank versus template-market term.**

Remark 11.3. Earlier papers sometimes treat weakwork as the dominant scalar rank. Later work correctly broadens it into one term in constrained template construction. The later interpretation is stronger and should supersede the earlier simplistic one.

11.4. **Unit economy as ontology versus accounting.**

Remark 11.4. The unit economy is now conceptually rich but only partially closed. The ontology is stronger than the accounting implementation. Future canonical drafts should finish the stock-flow and no-free-cycling layer.

11.5. **Trace-root semantics.**

Remark 11.5. The corpus has correctly noticed that state-root invariance under commutative swaps does not automatically imply trace-root invariance unless trace construction itself is canonicalized. This remains a subtle but important technical issue.

12. Distinctive Contributions of the Corpus

This section identifies the genuinely distinctive contributions of the internal literature.

12.1. **Pre-settlement market framing.**

Proposition 12.1 (Distinctive framing)**.** *The corpus's most distinctive framing move is to treat priced coordination before settlement as its own protocol object.*

Proof. Where many Bitcoin-adjacent systems optimize settlement throughput, execution flexibility, or payment efficiency, this corpus centers admission, replay, and checkpoint competition before settlement as the thing to be modeled and priced. □

12.2. **Candidate history rather than queue.**

Proposition 12.2 (History-over-queue contribution)**.** *The braid concept contributes a candidate-history model that is richer than ordinary queue or mempool analogies.*

12.3. **Constitutional kernel rather than broad VM.**

Proposition 12.3 (Constitutional-kernel contribution)**.** *The idea that the VM should be constitutional rather than general-purpose is one of the sharpest internal contributions of the corpus.*

12.4. **Multi-scarcity unit ontology.**

Proposition 12.4 (Economic contribution)**.** *The corpus contributes a multi-scarcity economic ontology rather than a fee-only or token-only model.*

12.5. **Productive hierarchy of agents.**

Proposition 12.5 (Agent-order contribution)**.** *The corpus contributes a real productive hierarchy of agents rather than flattening everything into generic users, miners, and apps.*

13. Weakest Points of the Corpus

13.1. **No frozen canonical specification.**

Remark 13.1. The corpus still lacks one fully authoritative frozen spec with byte layouts, activity schemas, anchor templates, and final fixtures.

13.2. **No working reference implementation.**

Remark 13.2. KapVM-Core has become spec-like, but there is still no reference interpreter or implementation results in the corpus itself.

13.3. **No empirical benchmark layer.**

Remark 13.3. There are still no measured replay costs, compression ratios, weakwork liquidity observations, treasury stress tests, or anchor fee distributions inside the corpus.

13.4. **Incomplete adversarial model.**

Remark 13.4. The adversarial layer remains lighter than the economic and conceptual layers. More work is needed on grinding, censorship, archive unavailability, knot instability, and cross-room or treasury-specific attacks.

13.5. **Overlapping document hierarchy.**

Remark 13.5. The corpus has reached the point where overlap itself is now a maintenance problem. A canonical ordering and pruning pass is necessary.

14. Proposed Canonical Condensation of the Corpus

This section proposes a clean six-document canon.

14.1. **Document I: Bitcoin as Extensible Consensus State.** This remains the foundation. It should be kept, tightened, and made clearly foundational rather than repeatedly expanded.

14.2. **Document II: Kapnet as a Pre-Settlement Coordination Market.** This document should carry weakwork, braids, replayability, knot logic, anchor theory, and mining implications.

14.3. **Document III: KapVM-Core Specification.** This should contain the machine model, state model, serialization rules, typed fragments, opcodes, errors, and test vectors.

14.4. **Document IV: Bitcoin Anchors, Witnesses, and Client Sync.** This should unify anchor classes, knot witness formats, fraud proofs, archive accountability, and light-client sync obligations.

14.5. **Document V: Treasury Accounting and Unit Economy.** This should close the economic and accounting layer: hedlbits, adjusted satdepth, hodlyield, utility balances, solvency, and no-free-cycling rules.

14.6. **Document VI: Institutions on the Amethyst State Chain.** This should unify submissions, rooms, guilds, TXXM, plugin ecology, daemon archetypes, agent-capable rooms, and governance logic.

Proposition 14.1 (Condensation benefit)**.** *A six-document canon of this form would preserve nearly all of the corpus's strongest ideas while reducing overlap and improving implementability.*

Proof. Each document would carry one principal burden: foundation, market theory, machine semantics, Bitcoin interface and witnesses, accounting, and institutional layer. This separates concerns while preserving structural continuity. □

15. A Canonical One-Sentence Thesis

Summary 15.1. The strongest one-sentence thesis now available for the corpus is:

Kapnet is a Bitcoin-native coordination substrate whose canonical objects are submissions, braids, knots, and anchors, and whose central novelty is the pricing and replay of institutional coordination before final settlement.

16. Main Synthesis

Theorem 16.1 (Corpus-synthesis theorem)**.** *The present Kapnet corpus constitutes a coherent internal body of work with the following structure:*

(i) a foundational theory of Bitcoin as extensible consensus state,
(ii) a market-theoretic model of pre-settlement coordination,
(iii) a political-economic model of agent hierarchy and multi-scarcity units,
(iv) an emerging protocol-kernel specification,
(v) a still-incomplete implementation, adversarial, and empirical layer.

Proof. The foundational documents define Bitcoin's role and Kapnet's admissibility as an extension layer. The market-theoretic documents define weakwork, braids, replayability, knots, and anchors. The economic documents define agent order, treasury continuity, and the unit ontology. The protocol-kernel documents define KapVM-Core, schemas, opcodes, witnesses, and test vectors. What remains underdeveloped are frozen implementation, adversarial proofs, and empirical benchmarks. Therefore the structure described above is an accurate synthesis. □

Corollary 16.2 (Maturity corollary)**.** *The corpus has completed its foundational phase, substantially advanced its market-theoretic and institutional phases, and begun but not completed its protocol and implementation phases.*

17. Conclusion

The Kapnet corpus has now matured to the point where it can no longer be understood as a loose set of speculative notes. It contains a real architecture of thought. That architecture begins with Bitcoin as final proof-of-work settlement, introduces Kapnet as a market for coordination before settlement, expands that

market into a political economy of differentiated agents and differentiated scarcities, and then begins to instantiate the resulting system as a bounded replay machine with typed institutional state.

This is a meaningful achievement. At the same time, the review makes clear that the corpus remains incomplete. It still requires a frozen reference specification, a reference interpreter, a stronger adversarial model, and empirical benchmarks. Yet incompleteness should not obscure the actual accomplishment. The corpus has already succeeded in one very hard task: it has identified a coherent and distinctive protocol space that is not reducible to ordinary L1s, ordinary L2s, or ordinary indexing systems.

Conclusion 17.1. Kapnet should now be understood as a Bitcoin-native pre-settlement coordination substrate in late theory-formation and early protocol-specification phase. Its foundational concepts are strong, its middle layers are unusually rich, its kernel is emerging, and its next requirement is disciplined condensation into a canonical spec-and-implementation program.

Acknowledgment. This synthesis is written in the spirit of treating the corpus itself as an object worthy of discipline, pruning, and preservation.

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